



Welcome to module 4

Hello again! You've come so far on your learning journey! Just think of all the new Python skills you've developed along the way. You've learned how to use variables to store and label your data, and how to work with different data types, such as integers, floats, and strings. You can call functions to perform actions on your data, and use operators to compare values. You also know how to write clean code that can be easily understood and reused by other data professionals. You can write conditional statements to tell the computer how to make decisions based on your instructions. And recently, you learned how to use loops to automate repetitive tasks. Coming up, we'll explore data structures, which are collections of data values or objects that contain different data types. Data professionals use data structures to store, access, organize, and categorize their data with speed and efficiency. Knowing which data structures fit your specific task is a key part of data work, and will help streamline your analysis. We'll focus on data structures that are among the most useful for data professionals: lists, tuples, dictionaries, sets, and arrays. Part of what makes Python such a powerful and versatile programming language are the libraries and packages that are available to it. After we review fundamental data structures, we'll discuss two of the most important libraries and packages for data professionals.

The first is Numerical Python, or NumPy, which is known for its high-performance computational power. Data professionals use NumPy to rapidly process large quantities of data. I often use NumPy in my job because it's so useful for analyzing large and complex datasets. The second is Python Data Analysis Library, or pandas, which is a key tool for advanced data analytics. Pandas makes analyzing data in the form of a table with rows and columns easier and more efficient, because it has tools specifically designed for the job. When you're ready, I'll meet you in the next video!

What you've learned

- Variables
- Data types
- Functions
- Operators
- Write clean code
- Conditional statements
- Loops

Data structures

Collections of data values or objects that contain different data types

What you'll learn

- Lists
- Tuples
- Dictionaries
- Sets
- Arrays
- NumPy
- pandas